

Thrust & Drift — Field Manual

Version 1.6.0 · Mongoose Traveller 2e Space Combat Simulator

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1. Overview

Thrust & Drift is a browser-based Virtual Tabletop (VTT) tool for running Mongoose Traveller 2nd Edition space combat at the gaming table. It is GM-operated and designed for shared-screen play — one person drives, everyone watches.

Two combat modes are available:

Mode	Description
VECTORIAL	Full hex grid with velocity vectors. Ships move according to their accumulated velocity each round. Implements <i>Traveller Companion 2024</i> pp. 169–186.
BASIC	Simplified range-band system. No hex map, no vectors. Faster for small engagements.

Note: The mode is selected on the Dashboard before starting a session and cannot be changed mid-battle.

2. Dashboard

The Dashboard is the pre-battle lobby. It has two columns: the **Operations Console** on the left and **Ship Profiles** on the right.

2.1 Ship Profiles

The right panel lists all saved ship profiles.

Control	Action
+ NEW PROFILE	Create a new ship from scratch.
 Edit	Modify an existing profile.
 Duplicate	Clone a profile as a starting point.
 Delete	Remove a profile (confirmation required).
 IMPORT	Load profiles from a <code>.json</code> file.
 EXPORT	Save all current profiles to a <code>.json</code> file.

Control	Action
 CATALOG	Browse the built-in High Guard 2022 catalog and add ships to your profiles.

2.2 Operations Console

Control	Action
COMBAT MODE	Toggle between Vectorial and Basic before starting.
 RESUME AUTOSAVE	Appears when an autosaved session is found. Shows round, phase, ship count, and timestamp. Click to restore instantly.
 NEW SESSION	Clears any existing battle state and enters the combat map.
 RESUME FROM FILE	Load a previously saved <code>.json</code> session. A preview screen shows the full roster before you confirm.
 FIELD MANUAL	Opens this manual inside the app.

2.3 Ship Profile Form

Each profile stores the ship's stats and named crew members.

Field	Description
Name / Tonnage	Display name and hull size (affects target size DM).
Hull / Thrust	Max hull points and base manoeuvre drive rating.
Turrets	Add weapon turrets: type, damage dice, range band, and special traits.

Field	Description
Crew	Named crew members, each with individual skill levels (see § 11).

3. Map Controls

The battle map is a **flat-top hex grid**. All interaction is mouse-driven.

Input	Action
Left-drag	Pan the map.
Scroll wheel	Zoom in / out.
Double-click	Centre the map on that hex.
Left-click token	Select the ship (highlights it).
Right-click hex	Open context menu — actions depend on the hex content and the current phase.

Note: Context menu actions are phase-gated. Only options valid for the current phase are shown. You cannot apply thrust during the Attack phase, for example.

4. Phase Flow

Each combat round follows this sequence. The HUD (top-left) shows the current round and phase. Click **NEXT PHASE** → to advance.

Phase	What happens
SETUP	Place ships on the map via right-click → <i>Add ship here</i> .
INITIATIVE	Roll initiative for all ships. Sets the acting order for the round.
ACCELERATION	Each ship applies thrust and optionally declares evasive action.
MOVEMENT	All ships move simultaneously according to their velocity vectors. (<i>Vectorial mode only — skipped in Basic mode.</i>)
ATTACK	Each ship in initiative order may attack or launch missiles.
ACTIONS	Each ship in initiative order may perform one crew action.
END OF ROUND	Round counter increments. Click NEXT PHASE to begin the next round.

5. Setup Phase

Place ships on the hex grid before the battle begins.

5.1 Adding a Ship

Right-click any empty hex → **Add ship here**. A modal opens where you select:

- **Profile** — which saved ship profile to use.
- **Faction** — Players, Allies, Enemies, Neutral. Affects token colour and auto-roll behaviour.
- **Color** — token display colour.

5.2 Removing a Ship

Right-click a ship token → **Remove from battle**. Available in all phases.

Note: Ships start with zero velocity. Their first Acceleration phase is used to build up speed.

6. Initiative Phase

Initiative determines the acting order for the Acceleration, Attack, and Actions phases.

Formula: `2D6 + Pilot skill + current Thrust rating + Tactics effect`
(MgT2e CRB p. 160)

6.1 Rolling Initiative

Right-click any hex → **Roll Initiative**. The modal shows all ships.

Ship type	Behaviour
Player ships (<code>faction =</code> <code>players</code>)	Dice inputs start empty . Enter the result of your physical 2D6 roll. The total updates live. CONFIRM is disabled until all player ships have entered dice.
NPC ships	Auto-rolled on confirm. Shown as  <i>auto</i> in the modal.
 button	Opt-in auto-roll — fills the dice fields for that ship if the player prefers the app to roll.

6.2 Tactics Bonus

If the Captain performs a Tactics check in the Actions phase of the previous round, the Effect is carried forward as a DM to the next round's initiative roll. The modal has a field to enter this bonus before rolling.

The Phase Tracker (right side of screen) shows the initiative order with the current actor highlighted.

7. Acceleration Phase

Each ship in initiative order adjusts its velocity vector and optionally declares evasive action.

7.1 Applying Thrust

Right-click ship → **Apply Thrust**.

The Thrust Modal shows the current velocity vector and a hex direction pad. Enter a delta-V (Δq / Δr) or click the direction buttons. The total magnitude of the delta cannot exceed the ship's available thrust (base thrust minus any M-Drive critical hit penalties).

A **ghost token** on the map previews where the ship will be next round if it maintains its current velocity after this thrust.

7.2 Evasive Action

Right-click ship → **Declare Evasion**. Allocate thrust points to evasion. These points are deducted from the available thrust for movement.

Evasive DM: $-(\text{Pilot skill} \times \text{evasive thrust points})$ (*MgT2e CRB p. 166*)

A **pulsing yellow aura** on the token indicates active evasion.

8. Movement Phase

Vectorial mode only. All ships move simultaneously — no player input required.

Click **NEXT PHASE** → to execute movement. Each ship's position advances by its current velocity vector.

8.1 Ships That Pass in the Night

If two hostile ships cross within **Short range (≤ 2 hexes)** during the same movement step — even if their final positions are far apart — the system detects the closest approach and opens the **Passing Encounter** window.

For each encounter the GM sees:

- Both ship names and faction colours
- Closest approach distance and range band
- **[Ship A] FIRES / [Ship B] FIRES** — opens the standard Attack Modal pre-set to that attacker; the GM can adjust weapon and target freely
- **PASS — LET THEM GO** — skips the opportunity with no attack

Multiple encounters are resolved sequentially; the window closes automatically when all are dismissed.

Note: Ships that end in the same hex trigger the **Dogfight** system instead and are handled at the movement → attack phase transition.

In Basic mode this phase is skipped automatically.

9. Attack Phase

Each ship in initiative order may make one attack or launch a missile salvo.

9.1 Attack Modal — Step 1: Config

Select the weapon to fire and the target ship. The modal shows a full DM breakdown:

- Range band DM
- Gunner skill DM
- Evasive action DM
- Sensor lock DM
- Weapon trait DM
- Target size DM
- Aid Gunners DM

9.2 Attack Modal — Step 2: Roll

Target number is always **8+** (*MgT2e CRB p. 164*).

Attacker	Behaviour
Player ships	Dice inputs start empty. Enter your physical 2D6 roll. CONFIRM ROLL disabled until both dice are entered.
NPC ships	Auto-roll button. Click to roll.

Effect = Total – 8. Positive effect = hit. Effect ≥ 6 = critical hit.

9.3 Attack Modal — Step 3: Damage

On a hit, roll damage dice for the weapon. Enter or confirm the damage roll. Damage is applied to the target's hull.

9.4 Attack Modal — Step 4: Critical Hit

Triggered automatically when Effect ≥ 6 , or when damage crosses a 10% hull threshold (Sustained Damage rule, *MgT2e CRB* p. 169).

Field	Description
Location	2D6 roll on the location table (Hull, M-Drive, J-Drive, Power Plant, Weapons, Sensors, Bridge, Fuel, Cargo, Crew, Computer).
Severity	Effect – 5, clamped 1–6. Stacks with existing criticals on the same system.
M-Drive	Sev 1 = no penalty. Sev 2–4 = –1 thrust/round. Sev 5–6 = thrust reduced to 0.

9.5 Launching Missiles

Right-click ship → **Launch Missiles** (Attack phase only). Select weapon and target. The missile salvo spawns as a token inheriting the launcher's velocity. Each round in the Movement phase it advances toward the target using its remaining thrust.

9.6 Sensor Lock

Acquired via the Sensors crew action. Grants a +DM to attacks against the locked target. Shown as an **animated cyan ring** on the locked ship.

9.7 Weapon Range Limits

Each weapon has a **maximum range band** beyond which it cannot fire (*MgT2e CRB* p. 167: "cannot attack targets beyond listed Range Band").

Weapon	Max Range
Railgun	Short
Beam Laser	Medium
Pulse Laser	Long

Weapon	Max Range
Particle Barbette	Very Long
Particle Beam	Very Long
Missile Rack	Special (no cap)
Sandcaster	Special (no cap)

When a target is beyond a weapon's max range:

- An **OUT OF RANGE** label appears on the weapon button in the Config step.
- If that weapon is selected, the error is shown above the roll button.
- **ROLL ATTACK is disabled** — the shot cannot proceed until a valid weapon or a closer target is chosen.

"Special" weapons (missiles, sandcasters) have no hard range cap and are never blocked by this rule.

10. Actions Phase — Crew

Each ship in initiative order may perform one crew action.

Right-click ship → **Crew Action**. The modal has three steps:

1. **Pick a crew member** — only members with relevant skills are shown.
2. **Pick an action** — filtered to the member's skill set.
3. **Roll** — if required (see below).

10.1 Skill DM Override

When an action is selected, the relevant skill level is pre-filled as the roll DM. The GM can override this value for specialisations (e.g. Engineer(M-Drive) 3 vs. generic Engineer 2). The  button resets to the base skill level.

10.2 Available Actions by Role

Captain

Action	Effect
TACTICS	2D6 + Leadership; Effect carried as initiative DM next round.
INSPIRE	Morale boost (descriptive, no roll).
COORDINATE	Aid another crew member's next action.

Engineer

Action	Effect
OVERLOAD DRIVE	Push thrust beyond rated maximum for one round (risk of damage).
REPAIR SYSTEM	Attempt to repair a critical hit on a specific system.
DAMAGE CONTROL	Reduce hull damage.

Gunner

Action	Effect
RELOAD	Reload a weapon that requires it.
AID GUNNERS	Provide a DM bonus to the next attack roll.

Sensors Operator

Action	Effect
SENSOR LOCK	Electronics(sensors) check; grants attack DM vs. locked target.

Action	Effect
ELECTRONIC WARFARE	Attempt to break an enemy sensor lock.

Note: Auto actions (some Captain actions) resolve immediately without a dice roll. Others require a 2D6 check. Player ships show dice input fields; NPC ships auto-roll.

11. Crew System

Ships have a list of **named crew members**, each with individual skill ratings.

11.1 Skills

Abbrev	Skill	Used for
PLT	Pilot	Initiative, evasion, thrust checks
CPT	Captain	Tactics, inspire, coordinate actions
ENG	Engineer	Drive overload, repair, damage control
GNR	Gunner	Attack DM, reload, aid gunners
SEN	Sensors	Sensor lock, electronic warfare

Skill levels range from **0 to 5**.

11.2 Multi-Skill Members

One crew member can hold **multiple skills**. A solo pilot/gunner on a fighter has both Pilot 2 and Gunner 2 on the same crew entry. The app resolves the highest relevant skill for each action automatically.

11.3 Editing Crew

In the ship profile form, use **+ ADD CREW** to add a member and **⊗** to remove one. Each row has a name field and five compact skill inputs (PLT / CPT / ENG / GNR / SEN).

Legacy profiles using the old flat object format (`{pilot: 2, gunner: 1}`) are automatically migrated to the named array format when loaded in the form.

12. Undo / Redo

Every user action that changes game state pushes a snapshot to the undo stack (capped at **20 entries**).

Control	Action
 Undo	Restore the previous state. Appears in HUD when stack is non-empty. Shortcut: <code>Ctrl+Z</code> / <code>Cmd+Z</code> .
 Redo	Re-apply an undone action. Appears in HUD when redo stack is non-empty. Shortcut: <code>Ctrl+Y</code> / <code>Cmd+Shift+Z</code> .

Both buttons are hidden when their respective stacks are empty — they appear only when relevant.

*The battle log is **not** rolled back on undo. Instead, a  **Undo** entry is appended to the log so the action history remains readable.*

13. Save & Resume

13.1 Autosave

The app autosaves to IndexedDB after every significant action (ships added/removed, damage applied, phase advanced, etc.). No manual trigger needed.

On next visit, the **⏮ RESUME AUTOSAVE** button appears on the Dashboard with round, phase, and ship count.

13.2 Manual Save

Click  **SAVE** in the HUD at any time to download the full session as a `.json` file.

13.3 Resume from File

On the Dashboard, click ↓ **RESUME FROM FILE** and select your saved `.json` file. A preview screen shows the full roster (name, faction, hull, position) before you confirm loading.

13.4 Profile Export / Import

Ship profiles are separate from battle sessions. Use ↑ **EXPORT** and ↓ **IMPORT** in the profile panel to share or back up profiles independently.

Clicking  in the HUD returns to the Dashboard. A confirmation modal warns that unsaved battle data will be lost — save first if you need to resume.

14. Dogfight

MgT2e CRB p.138 — Vehicle Combat / Dogfighting rules adapted for space combat.

14.1 What is a Dogfight?

A **dogfight** is a close-range sub-system that activates automatically when two or more ships from different factions end the **Movement phase in the same hex** (vectorial mode only).

Standard combat is suspended for the involved ships. Instead, the round subdivides into **6 micro-rounds** of 6 seconds each — one full standard round equals six micro-rounds of dogfight.

14.2 Engagement

When the app detects a potential dogfight at the end of Movement, the **CONFIRM INTENTS** modal opens.

For each ship the GM declares:

Intent	Outcome
Both YES	Dogfight activates immediately
Both NO	Ships treated as Short Range (distance 1) — no dogfight
Mixed	Pursuit check required (see §14.3)

14.3 Pursuit Check

Formula: **2D6 + Pilot + Tonnage DM + free Thrust**

- **Free Thrust** = profile thrust – thrust used this round
- **Tonnage DM**: <50t → 0; 50–99t → –1; 100–199t → –2; +–1 per 100t above 100

If the pursuer's total exceeds the evader's total → dogfight activates. Otherwise → Short Range, no dogfight.

Enter dice manually for each side. The app computes and compares totals live.

14.4 Micro-Round Flow

Open the round from the **HUD dogfight tracker** (✕ DOGFIGHT panel, top-left).

Step 1 — Declare escape (optional)

At the start of each micro-round the GM may declare that a ship wants to flee.

- If its thrust exceeds all enemy thrusts → **auto-escape**, no check needed.
- If enemies choose not to pursue → **auto-escape** (toggle "NOT PURSUING").
- Otherwise → pursuit check (same formula as §14.3).

Step 2 — Pilot check

Each remaining ship rolls **2D6 + Pilot + Tonnage DM + Thrust + previous round bonus**.

DM	Source
Pilot skill	<code>getCrewSkill(crew, 'pilot')</code>
Tonnage DM	See §14.3 table
Thrust	Profile thrust – thrust used this round
Extra enemies	–(number of enemy ships – 1) when outnumbered
Round bonus	Previous round winner's margin carries forward as a +DM

Step 3 — Result

Outcome	Effect
Winner	+2 DM to all attacks this micro-round; chooses enemy fire arc
Loser	–2 DM to all attacks
Tie	Fixed weapons cannot fire; turrets OK; no positional DM

Apply the attack DMs shown in the modal when opening the **Attack** panel.

Step 4 — Advance

Click **ADVANCE** → **MICRO-ROUND N+1/6**. After micro-round 6 the dogfight ends automatically and all ships return to normal combat flow.

14.5 Token Visuals

Ships in a dogfight display:

- **Pulsing amber ring** around the token
- **✕ badge** top-right of the token
- Ghost position and velocity arrow are hidden (no movement during dogfight)

14.6 Escape Mid-Dogfight

Escape can be declared at the start of any micro-round (Step 1 above). Conditions:

```
Auto-escape: ship.thrust > max(enemy thrusts)
              OR enemies choose not to pursue
Check:       2D6 + Pilot + Tonnage DM + free Thrust (same as §14.3)
              evader total > pursuer total → escaped
```

On successful escape `inDogfight` is cleared; the ship re-enters normal combat from the next standard round.

15. Boarding

HG 2022 pp.125–135 — full 4-phase boarding system.

Boarding is a sub-system that activates when an attacker moves adjacent to a target and meets the thrust requirement. Combat shifts from the hex map to the interior of the target ship.

15.1 Triggering a Boarding Action

Right-click the **attacker ship** and select ✕ **Board [target name]...**

The option is visible only when:

```
distance(attacker, target) ≤ 1 (Adjacent or Close)
AND attacker.thrust ≥ target.thrust
    OR target M-Drive critical is disabled
AND different factions
```

The **Boarding Setup** modal opens. Select the target and confirm. The boarding moves immediately to Phase 2 — Contact.

15.2 Phase 1 — Approach

Handled outside the app. The approach phase is the normal combat movement that brought the two ships together. The GM declares the boarding when conditions are met.

Voluntary boarding (target cooperates): skip to Contact immediately.

Forced boarding: if the target attempted to flee and failed (or is immobilised), proceed to Contact.

15.3 Phase 2 — Contact

The ✕ **CONTACT** modal opens. The GM selects the entry method:

Method	Check	Difficulty	Time	DM
Airlock (cooperative)	None	—	Instant	—
Airlock (forced)	Mechanic (STR)	14+	2D rounds + 1D	—
Maintenance Hatch	Mechanic (STR)	12+	2D rounds	— ⚠
Breaching Tube	None	—	< 2 min	—
Forced Linkage Apparatus	Pilot (DEX)	8+	Immediate	+2
Hull Cut	Mechanic (DEX)	8+/round	Per round	— ⚠

⚠ = decompression risk if compartment not evacuated.

Modifiers:

- 🌀 **Tumbling** — defender rotating the ship: DM -1 to all Contact checks
- 🔗 **Forced Linkage** — DM +2 to all Contact checks; defender cannot manoeuvre

Hull Cut tracker: select component (Hatch / Airlock / Hull) and cutting tool, roll each round. Damage reduces component Resilience; breach achieved when damage ≥ breach threshold.

When entry is secured, click **ADVANCE TO CONFLICT** → .

15.4 Phase 3 — Conflict

The ✕ **CONFLICT** modal tracks the boarding fight.

Tactical objectives — check each when captured:

Objective	Effect when captured
Bridge	Remote control of all ship systems disabled for enemy
Engineering	Propulsion, reactor, life support under attacker control
Turrets	Weapon systems under attacker control

The ship is considered taken when all three objectives are captured.

Combat tools:

- **ROLL STACKING** — roll 2D ≥ 10 to target a combatant beyond the first in a corridor (HG p.131)
- **ROLL MISSED SHOT** — roll 2D on the missed-shot table for every attack that misses; optional **Armored bulkhead (DM -1)** toggle

Weapon DM in tight spaces: Rifles -2 · Heavy weapons -4 · Grenades → automatic 6D+

When the fight is resolved, click **END CONFLICT — ADVANCE TO SECURITY** → .

15.5 Phase 4 — Security

The ✕ **SECURITY — BOARDING OUTCOME** modal resolves the action.

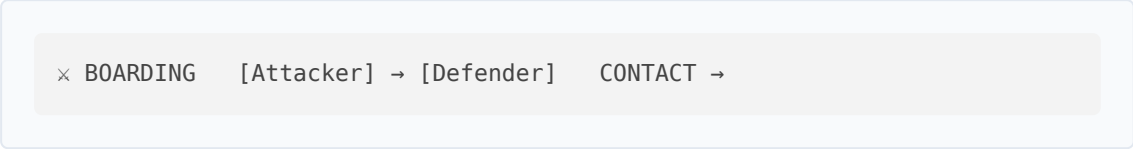
Outcome	Effect
Attacker wins	Boarding party controls the ship; optional faction transfer
Defender repels	Boarders eliminated, captured, or driven off
Ship destroyed	Target destroyed by internal damage during conflict

If **Attacker wins** and faction transfer is enabled, the captured ship's faction changes to match the attacker. Enemy crew is removed from the roster.

Click **CONFIRM OUTCOME** to close the boarding.

15.6 HUD Indicator

While a boarding is active, the HUD shows a ✕ **BOARDING** badge below the standard tracker:

A screenshot of a HUD indicator. It consists of a light blue rounded rectangle containing a light gray rounded rectangle. Inside the gray rectangle, the text '✕ BOARDING' is on the left, followed by '[Attacker] → [Defender]' in the center, and 'CONTACT →' on the right.

✕ BOARDING [Attacker] → [Defender] CONTACT →

Click the phase button to reopen the relevant modal at any time.

15.7 Boarding and Normal Combat

- Ships **inBoarding** do not participate in the standard Attack phase
- A ship with **Forced Linkage active** cannot use thrust to manoeuvre
- If the target is destroyed during Conflict, resolve with outcome **Ship destroyed**
- Normal rounds continue in parallel — the GM can advance phases and resolve the boarding on its own timeline, as with dogfights

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